























# Technical Architecture

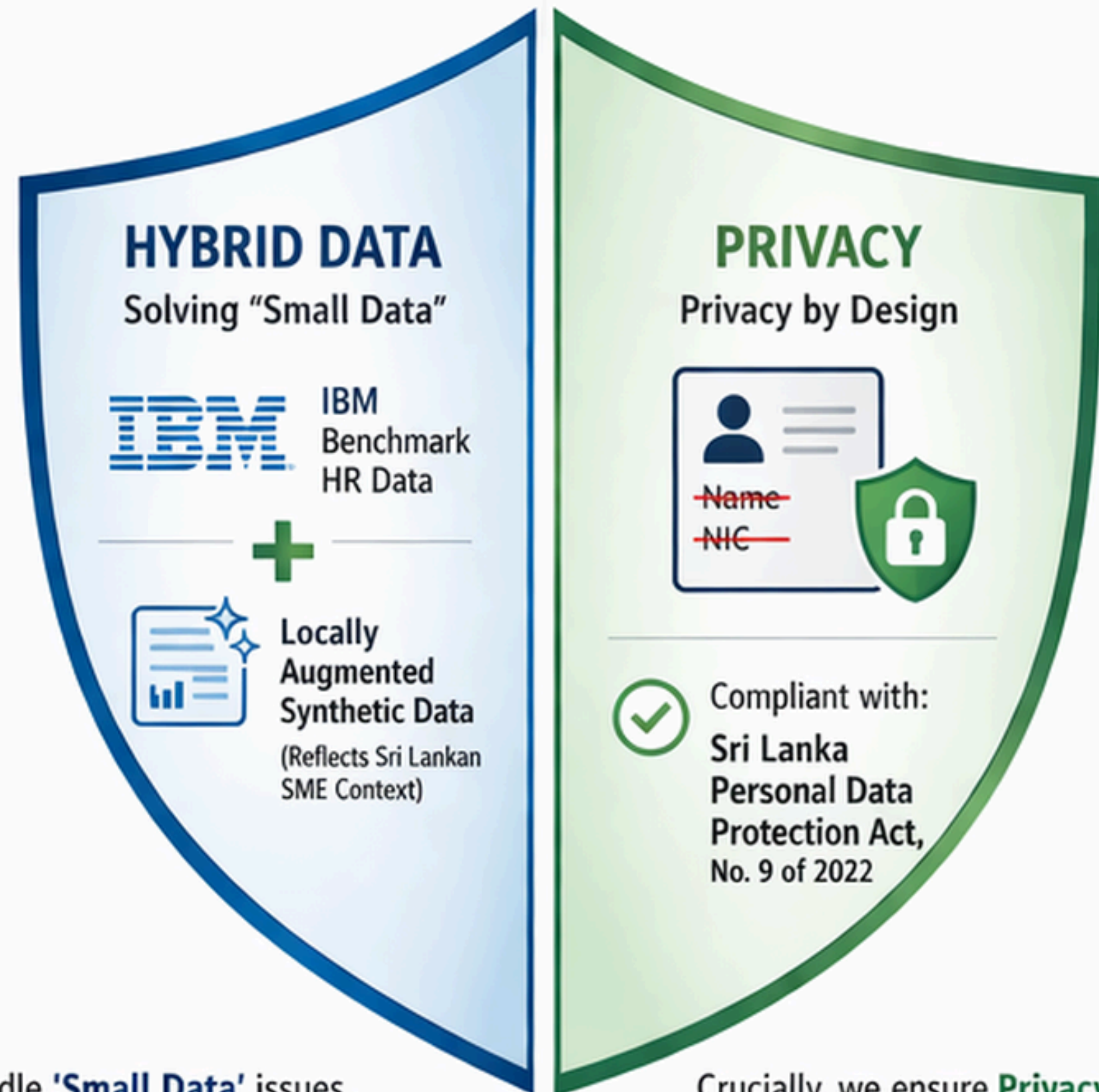


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*Serverless Batch Prediction Pipeline:* Cloud Function triggers at midnight, calls Vertex AI, processes the data, and stores the results in the database. Scales to Zero.

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# Data Strategy & Privacy



To handle '**Small Data**' issues, we use a hybrid approach: **IBM's benchmark data** for patterns and **locally augmented synthetic** data for context.

Crucially, we ensure **Privacy by Design**. All names and NICs are stripped before the AI sees the data, complying with the **Sri Lanka Personal Data Protection Act No. 9 of 2022**.



# Performance & Success Metrics

Success will be measured by three pillars: **Technical accuracy** (prioritizing Recall to minimize expensive false negatives), **User satisfaction** (using the System Usability Scale), and **Financial viability** (a strict audit of Google Cloud billing logs).



High Technical Accuracy

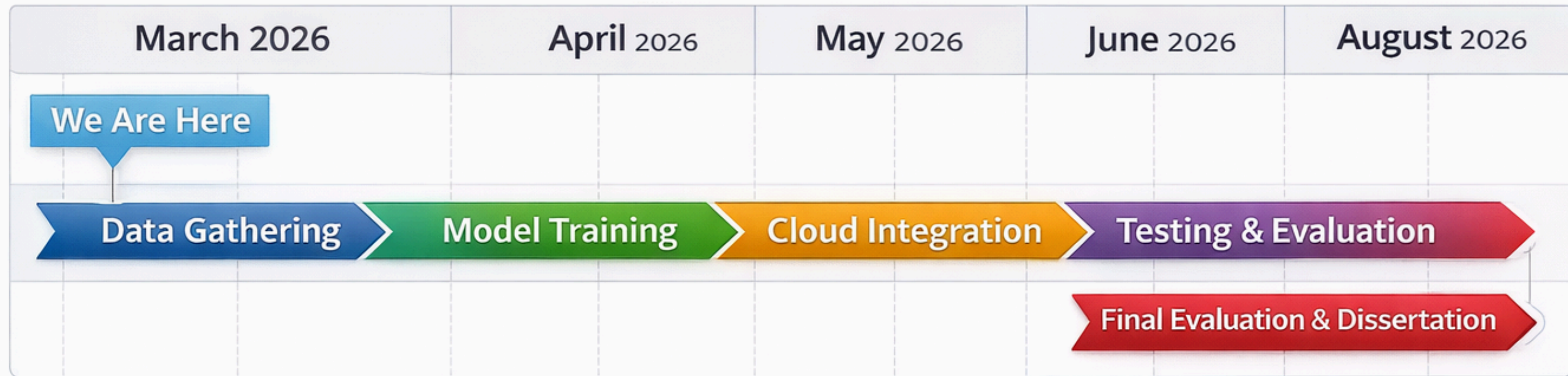


High User Satisfaction



Financial Viability

# Timeline



■ Data Gathering   ■ Model Training   ■ Cloud Integration   ■ Testing & Evaluation   ■ Final Evaluation & Dissertation



# In Conclusion

NexusHR isn't just a software module; it's a proof-of-concept for how Sri Lankan SMEs can use cutting-edge AI to fight the Brain Drain **without breaking the bank.**

## Proactive Retention: Fighting the Brain Drain

### Democratization: AI for Every SME



Affordable AI solutions  
for all small and medium enterprises.

### Scalability: Beyond Sri Lanka



A model that can expand  
to the global market.

### Proactive Retention: Fighting the Brain Drain



Keeping top talent and  
reducing employee turnover.





**THANK YOU**

